

AI-POWERED HEALTHCARE PLATFORM

ArogyaAI

From patient symptoms to doctor-ready clinical briefs —
in any language, before the consultation begins.

Team Name • Hackathon 2026

THE PROBLEM

Meet Ramesh

*"Ramesh is 58. He carried a folder of lab reports to a government hospital. He waited **3 hours**. His consultation lasted **90 seconds**. The doctor never opened his folder. Ramesh has undiagnosed diabetes."*

This isn't rare. This is the norm.



Under 2 minutes — average consultation time in India^[1]



Paper reports ignored — doctors have no time to read unstructured documents



Language barrier — most Indians don't speak English^[est.], yet health AI is English-only

1:1,457

Doctor-patient ratio
WHO recommends 1:1,000^[2]

~63%

Out-of-pocket health spending
NHA 2019–20^[3]

22

Official languages in India
Health AI speaks 1

OUR SOLUTION

ArogyaAI — Making Every 2 Minutes Count

An end-to-end AI platform that speaks the patient's language, understands their documents, and prepares the doctor — before the consultation begins.



Speak Your Language

Voice input in Hindi, Marathi, English — even mixed Hinglish



AI Understands Reports

Lab values extracted with AI self-assessment indicators you can verify



Doctor Gets a Brief

Structured clinical copilot with urgency, conditions, and focus areas

💡 **Key:** These aren't separate apps. It's one integrated pipeline — symptoms → documents → booking → AI copilot brief → doctor.

HOW IT WORKS

The Patient Journey — End to End



Describe Symptoms

Voice or text
Hindi / Marathi / English



Upload Reports

PDF / image
Lab reports, prescriptions



AI Analyzes

OCR + Gemini
AI quality indicators



Patient Verifies

Review & correct
Human-in-the-loop



Doctor Gets Brief

AI Copilot output
Urgency + conditions



Screenshot: Patient Dashboard

Capture from /dashboard — show the
health records + appointments overview



Screenshot: Doctor Dashboard

Capture from /doctor/dashboard — show
today's appointments with AI brief badges

AI Pre-Consultation Copilot

Before the appointment, AI synthesizes symptoms + reports into a **structured clinical brief** the doctor reviews before the patient walks in.



Accepts symptoms in Hindi, Marathi, English



Correlates with uploaded report data



Assigns urgency: LOW → CRITICAL



Suggests focus areas + recommended specialty



Flags abnormal lab values with normal ranges



Analysis reliability indicator (HIGH / MEDIUM / LOW)

⚠ Advisory output only — not a clinical diagnosis. Always reviewed by the consulting physician.

```
{
  "summaryText": "Patient reports fever and headache for 3 days. Lab reports indicate low hemoglobin",
  "symptoms": ["fever", "headache", "fatigue"],
  "symptomTimeline": "3 days",

  "urgencyLevel": "MEDIUM",
  "conditions": ["Anemia", "Viral fever"],

  "suggestedFocusAreas": [
    "Complete blood count",
    "Iron studies"
  ],
  "abnormalValues": [{
    "parameter": "Hemoglobin",
    "value": "8.2 g/dL",
    "normalRange": "12.0-17.5 g/dL"
  }],
  "analysisReliability": "HIGH",
  "recommendedSpecialty": "General Physician"
}
```

Medical Document Intelligence



● High — AI Self-Assessment

Score ≥ 0.8 — AI reports high extraction certainty

● Medium — Verify Recommended

Score 0.5–0.8 — patient should double-check this value

● Low — Manual Check Needed

Score < 0.5 — patient should enter the correct value



Screenshot: Verification Modal

Capture the MedicalVerificationModal showing green/amber/red indicators on extracted values

Honest note: These indicators are AI-generated, not clinically validated against ground truth. They serve as *attention signals* — helping patients identify which values to double-check.

📄 Patient corrections are tracked for future model improvement.

Multilingual **Voice-First** Input

Patients describe symptoms by **speaking naturally** — in their own language. The AI mirrors the language and normalizes for doctors.

GB ENGLISH

IN हिंदी

IN मराठी



Real-time live transcription (browser-native speech recognition)



AI mirrors input language in responses



Urgency keywords (chest pain, breathlessness) trigger instant alerts



Quick-pick symptom tags for common conditions



Hinglish (mixed Hindi-English) handled naturally



Voice Input (Hindi):

"मुझे 3 दिन से बुखार और सिरदर्द है, और सीने में दर्द हो रहा है"

▲ **Urgent symptom detected:** chest pain — call 112 if emergency

AI normalizes for doctor's brief:

→ fever, headache, chest pain (3-day duration)

AI-Powered Patient 360°

When the doctor opens a patient profile, they see an AI Command Center — not a blank chart.



AI Patient Snapshot

Risk badge (LOW→CRITICAL), current symptoms, conditions, allergies — one hero view with the AI pre-consultation brief



Consultation Prep Panel

AI-suggested focus areas, new symptoms since last visit, resolved symptoms, recent abnormal findings



Health Trends

Biomarker trend charts (hemoglobin, glucose, cholesterol, etc.) with SVG line charts showing history



Health Timeline

Chronological patient history — appointments, reports, risk levels — with inline document viewer



Uploaded Reports Viewer

View PDFs and images inline. See AI analysis severity and extraction data for each report.



Emergency Triage

Patient-facing emergency page with 4-level AI severity classification (CRITICAL/HIGH/MODERATE/LOW) and 112 quick-dial



Screenshot: PatientView360 — AI Snapshot Hero

Capture `/doctor/patients/id/360` showing the risk badge, symptoms, conditions, and AI brief summary. This is your most impressive UI.

Technical Foundation

Frontend

React 18 + Vite
Tailwind CSS + Framer Motion
Zustand state management
Browser-native voice API
Lazy-loaded routes

Backend

Node.js + Express.js
Modular service layer
JWT + OTP authentication
Cron-based reminders
Role-based access (patient/doctor)

AI & Data

Google Gemini API
4-model fallback chain
MongoDB Atlas
Cloudinary CDN
Nodemailer (email)

AI Resilience

4-tier model fallback: gemini-flash-latest → 2.0-flash-lite → 2.0-flash → 2.5-flash. Daily quota exhaustion is cached — no wasted API calls. Per-minute rate limits use server-suggested retry delays. Safety blocks throw immediately, never retry.

Security

Helmet.js security headers, MongoDB injection sanitization, HTTP parameter pollution prevention, CORS origin whitelist, role-based rate limiting (auth: 20/15min, AI: 20/15min, uploads: 10/15min), medical disclaimers on all AI outputs.

Competitive Landscape

AI CAPABILITY	PRACTO	ADA HEALTH	NUANCE DAX	AROGYAAI
Appointment Booking	✓	—	—	✓
AI Symptom Chat	—	✓	—	✓
Medical Document AI Analysis	—	—	—	✓
AI Quality Indicators on Extraction	—	—	—	✓
Human-in-the-Loop Verification	—	—	—	✓
Pre-Consultation Copilot	—	—	Post-visit	✓
Multilingual Voice (Hindi/Marathi)	—	—	EN only	✓
AI Emergency Triage	—	✓	—	✓
Total AI Features	1/8	2/8	1/8	8/8

⚡ **Honest note:** This comparison focuses on AI capabilities. Practo has millions of users, real payments, and verified doctors — operational capabilities we haven't built yet. Our advantage is in AI intelligence and integration, not operational scale.

Who Benefits — And Why We Can Win

For Patients

 **Accessibility:** Speak symptoms in your own language

 **Transparency:** See AI extractions, correct mistakes

 **Continuity:** Health records organized, follow you

 **Safety:** Emergency triage when seconds matter

For Doctors

 **Pre-read Brief:** Structured AI copilot before consultation

 **Time Efficiency:** Even 1 min saved per patient compounds across 40+ daily consultations

 **Structured Data:** Abnormal values flagged, trends charted

 **Automation:** Cron-based appointment reminders reduce no-shows

Responsible AI

 **Advisory only** — not a diagnosis or medical device

 **Human verification** for all extracted medical data

 **Doctors remain** the final decision makers

 **Medical disclaimers** on every AI output

 **Working Prototype**
Running code, not slides

 **5 AI Capabilities**
Chat, OCR, copilot, triage, summary

 **3 Languages**
Hindi, Marathi, English

 **Deployment-Ready**
Fallback chain, rate limiting, auth

 **1.4B People**
Solving a real problem at scale

Future Scope & Validation Roadmap

Future Development Phases

PHASE 1 NOT YET IMPLEMENTED

Verified doctor onboarding · Razorpay payments · Video consultations

PHASE 2 NOT YET IMPLEMENTED

ABDM / ABHA integration · Digital prescriptions · Unified health records

PHASE 3 NOT YET IMPLEMENTED

Chronic disease prediction · Longitudinal health monitoring · AI visit summaries

PHASE 4 NOT YET IMPLEMENTED

Android & iOS apps · Tamil, Telugu, Bengali, Kannada

PHASE 5 NOT YET IMPLEMENTED

Wearable integrations · Remote patient monitoring

Validation & Ethics Roadmap

 **Clinical validation** with practicing doctors

 **Real-world pilot testing** in healthcare facilities

 **Accuracy benchmarking** against ground truth data

 **Explainability improvements** for AI decisions

 **Bias and safety evaluation** across demographics

 **Regulatory readiness** assessment (DISHA, CDSCO)

⚠ All validation items are **future work**. We know the path from prototype to product — and we've designed the architecture to support it.

Thank You

In India, a doctor sees 40 patients a day.

Each consultation lasts 2 minutes.

We can't add more minutes.

But we can make every minute count.



GitHub: [ArogyaAI](#)



Live Demo Available

Questions? We'd love to discuss the technical details — or show a live demo.